

Can Perform Polynomial Division

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July 1, 2026

Introduction

Polynomial division is a process similar to the long division of numbers. It allows us to divide one polynomial (the dividend) by another polynomial (the divisor) to obtain a quotient and a remainder. Polynomial division is particularly useful when simplifying expressions or solving polynomial equations.

Step-by-Step Process

To divide one polynomial by another, follow these steps:

1. **Arrange the Polynomials:** Write the dividend and divisor polynomials in descending order of their degrees.
2. **Divide the Leading Terms:** Divide the leading term (the term with the highest degree) of the dividend by the leading term of the divisor to get the first term of the quotient.
3. **Multiply and Subtract:** Multiply the entire divisor by the term obtained in step 2, and subtract the result from the dividend.
4. **Repeat:** Repeat the process with the new polynomial obtained after subtraction until the degree of the remainder is less than the degree of the divisor.

Example 1: Simple Polynomial Division

Divide $2x^3 + 3x^2 + 4x + 5$ by $x + 1$.

Solution:

$$\text{Step 1: } \frac{2x^3 + 3x^2 + 4x + 5}{x + 1}$$

$$\text{Step 2: } \frac{2x^3}{x} = 2x^2$$

$$\begin{aligned}\text{Step 3: } (2x^3 + 3x^2 + 4x + 5) - (2x^2 \cdot (x + 1)) \\ = 2x^3 + 3x^2 + 4x + 5 - (2x^3 + 2x^2) = x^2 + 4x + 5\end{aligned}$$

$$\text{Step 4: } \frac{x^2}{x} = x$$

$$\text{Step 5: } (x^2 + 4x + 5) - (x \cdot (x + 1))$$

Example 1: Simple Polynomial Division

Solution Continued:

$$\text{Step 6: } \frac{3x}{x} = 3$$

$$\begin{aligned}\text{Step 7: } & (3x + 5) - (3 \cdot (x + 1)) \\ & = 3x + 5 - (3x + 3) = 2\end{aligned}$$

Quotient: $2x^2 + x + 3$

Remainder: 2

Example 2: Division a Higher-Degree Dividend

Divide $x^4 - 3x^3 + 5x^2 - x + 2$ by $x - 1$.

Solution:

$$\text{Step 1: } \frac{x^4 - 3x^3 + 5x^2 - x + 2}{x - 1}$$

$$\text{Step 2: } \frac{x^4}{x} = x^3$$

$$\begin{aligned}\text{Step 3: } & (x^4 - 3x^3 + 5x^2 - x + 2) - (x^3 \cdot (x - 1)) \\ & = x^4 - 3x^3 + 5x^2 - x + 2 - (x^4 - x^3) \\ & = -2x^3 + 5x^2 - x + 2\end{aligned}$$

$$\text{Step 4: } \frac{-2x^3}{x} = -2x^2$$

$$\begin{aligned}\text{Step 5: } & (-2x^3 + 5x^2 - x + 2) - (-2x^2 \cdot (x - 1)) \\ & = -2x^3 + 5x^2 - x + 2 - (-2x^3 + 2x^2) = 3x^2 - x + 2\end{aligned}$$

Example 2: Division a Higher-Degree Dividend

Solution Continued:

$$\text{Step 6: } \frac{3x^2}{x} = 3x$$

$$\begin{aligned}\text{Step 7: } (3x^2 - x + 2) - (3x \cdot (x - 1)) \\ = 3x^2 - x + 2 - (3x^2 - 3x) = 2x + 2\end{aligned}$$

$$\text{Step 8: } \frac{2x}{x} = 2$$

$$\begin{aligned}\text{Step 9: } (2x + 2) - (2 \cdot (x - 1)) \\ = 2x + 2 - (2x - 2) = 4\end{aligned}$$

Quotient: $x^3 - 2x^2 + 3x + 2$

Remainder: 4

Example 3: Division results in a Zero Remainder

Divide $x^2 - 6x + 9$ by $x - 3$.

Solution:

$$\text{Step 1: } \frac{x^2 - 6x + 9}{x - 3}$$

$$\text{Step 2: } \frac{x^2}{x} = x$$

$$\begin{aligned}\text{Step 3: } & (x^2 - 6x + 9) - (x \cdot (x - 3)) \\ & = x^2 - 6x + 9 - (x^2 - 3x) = -3x + 9\end{aligned}$$

$$\text{Step 4: } \frac{-3x}{x} = -3$$

$$\begin{aligned}\text{Step 5: } & (-3x + 9) - (-3 \cdot (x - 3)) \\ & = -3x + 9 - (-3x + 9) = 0\end{aligned}$$

Quotient: $x - 3$

Remainder: 0

Conclusion

Polynomial division is a fundamental algebraic technique that is essential for simplifying expressions and solving equations. By following the step-by-step process outlined above, you can successfully divide any polynomial by another, producing a quotient and remainder.